Three-Dimensional Frequency-Domain Transform Network for Cross-Scene Hyperspectral Image Classification

Jun Zhang, Cheng Zhang, Shuai Liu, Zhenwei Shi, *Senior Member, IEEE*, and Bin Pan, *Member, IEEE*

***Abstract*— Reducing interdomain discrepancies effectively enhances the performance of hyperspectral cross-scene classification tasks. However, hyperspectral single-source domain (SD) generalization methods based on mining visual representation information are significantly influenced by interdomain discrepancies. Recent research has demonstrated that frequency-domain information exhibits robust stability. Therefore, this article proposes a three-dimensional frequency domain transform network (TFTnet) for achieving hyperspectral single-SD cross-scene classification tasks. To leverage the advantageous 3-D characteristics of hyperspectral images (HSIs), all frequency domain transforms are implemented within a 3-D framework. The model consists of a generator and a discriminator. The generator incorporates a frequency domain enhancement (FDE) module and a multisource information fusion (MIF) module; the discriminator incorporates a set of weight-sharing adaptive frequency domain transform (AFT) modules. The FDE module generates the extended domain (ED) with a certain domain shift by doing linear interpolation in the amplitude interval of a single SD itself. The MIF module integrates multisource information through interdomain attention, ensuring a balanced approach between the SD and ED, thus generating the effective balance domain (BD). The AFT module empowers the discriminator to selectively acquire HSI frequency domain features, facilitating synergistic collaboration of spatial–spectral features and frequency domain features for enhanced image comprehension. Extensive experiments on three public hyperspectral datasets show the superiority of the method compared with state-of-the-art techniques.**

***Index Terms*— Cross-scene classification, interdomain discrepancies, three-dimensional frequency domain transform.**

I. Introduction

HYPERSPECTRAL image (HSI) has abundant image information, and its classification task based on convolutional neural network has achieved effective results in identifying land cover information accurately, which has a great practical value [1], [2], [3], [4], [5], [6], [7], [8]. However, in practical applications, due to unpredictable real-world factors, a domain shift exists between the training images and target images. This shift causes the trained model to not effectively recognize the new target domain (TD), leading to a decrease in the performance of the classification model [9], [10], [11].

In HSI cross-scene classification tasks, the domain shift problem has been effectively addressed by domain adaptation methods [12], [13], [14], [15], [16], [17], [18]. Domain adaptation aims to train the model with labeled source domain (SD) data and partially unlabeled TD data and then extract domain-invariant features [19]. However, TD data cannot always be seen in actual training. Therefore, the task setting of learning domain invariant features and reducing inter-domain variability is more in line with practical needs when training with SD data only, this task setting is more in line with the practical needs.

Domain generalization (DG) shows promising effects in addressing the performance degradation of models caused by interdomain discrepancies in HSI cross-scene classification tasks. DG effectively generalizes from one or more known SDs to an unknown TD by reducing the interdomain distance between single or multiple SDs [20], [21], [22], [23], [24], [25] and trains models that are more adaptive to domain variations [26], [27]. Recent research for the HSI cross-scene classification task is mainly based on single-SD generalization, where only data from one SD are used to train the model. The current common single-source DG solution is to construct simulated samples, expand interdomain relations, and then acquire domain-invariant features by methods, such as contrast-adversarial learning. Decomposing the spatial and spectral information of images, reconstructing the samples at the visual level [28], [29], [30], and realizing domain expansion are effective sample expansion methods. Wang et al. [31] propose a domain expansion network that adopts adversarial contrastive methods to learn domain-invariant features and generate augmented samples using spatial randomization and spectral randomization. Dong et al. [32] propose a spatial–spectral augmentation and causal constraint network in terms of data augmentation and causal alignment and construct a causal contribution discriminator. Some studies believe that mining linguistic textual knowledge is also an effective way to make up for the lack of sample size.

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Jun Zhang, Cheng Zhang, and Shuai Liu are with the School of Artificial Intelligence, Hebei University of Technology, Tianjin 300401, China, and also with Hebei Province Key Laboratory of Big Data Calculation, Tianjin 300401, China (e-mail: zhangjun@scse.hebut.edu.cn).

Zhenwei Shi is with the Image Processing Center, School of Astronautics, and the State Key Laboratory of Virtual Reality Technology and Systems, Beihang University, Beijing 100191, China (e-mail: shizhenwei@buaa.edu.cn).

Bin Pan is with the School of Statistics and Data Science, KLMDASR, LEBPS, and LPMC, Nankai University, Tianjin 300071, China (e-mail: panbin@nankai.edu.cn).

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Zhang et al. [33] propose a language-aware DG network, which utilizes linguistic features as a cross-domain shared semantic space and completes the visual linguistic alignment through supervised contrastive learning of the semantic space.

However, current DG research for HSI only considers visible visual representation information, aiming to learn domain-invariant representations at the spatial–spectral level, while neglecting the invisible frequency-domain information.

Recent methods show that frequency domain information is not susceptible to domain shifts [34], [35], [36], [37], [38], which is crucial in cross-scene tasks. Incorporating frequency-domain information into HSI cross-scene tasks aims to achieve more stable domain-invariant representations. Therefore, this article hopes to incorporate frequency-domain information into HSI cross-scene classification tasks.

In this article, a three-dimensional frequency domain transform network (TFTnet) is proposed for realizing the single-SD cross-scene classification task of HSI by combining the favorable 3-D characteristics of HSI. The framework fully considers the HSI image data characteristics and places all the frequency domain transformations on the spatial and spectral domains, and the frequency domain operations get complete in the 3-D spatial domain. TFTnet consists of two main components: a generator incorporating a frequency-domain enhancement module and a multisource information fusion (MIF) module, used to generate extended domains (EDs) with specific frequency-domain features; and a discriminator containing an adaptive frequency-domain transformation module, designed for spatial-frequency coordinated classification tasks. Specifically, in the generator, ED and balance domain (BD) with certain domain shifts generate by linearly interpolating the amplitudes in the domain and applying weighted constraints on the multisource information, and the two domains together form the extended samples to help the model achieve better generalization performance. In the discriminator, autonomous selection of frequency domain transformations carries out to achieve spatial-frequency synergy, further increasing the model's learnable information.

The main contributions of this work summarize as follows.

1) The proposed TFTnet applies frequency domain transformation to HSI cross-scene classification tasks, achieving a 3-D transformation from spatial–spectral to frequency domain information.
2) The developed frequency domain enhancement (FDE) module and MIF module form the generator, facilitating intradomain and interdomain information interaction, generating more representative frequency domain information and ensuring the reliability of generated samples.
3) The developed adaptive frequency domain transform (AFT) module achieves spatial-frequency synergy, allowing the discriminator to autonomously learn more information about the frequency domain.

The rest of this article is organized as follows. Section II describes the proposed TFTnet. Section III presents the experimental content and result analysis of this work. Finally, Section IV draws the conclusions.

II. Methodology

A. Background

The goal of single-source DG is to use data from a single SD to generate one or more EDs with a certain domain bias and then train more adaptive models that can perform cross-scenario tasks well by reducing interdomain distributional discrepancies and learning intradomain knowledge

$$X_{\mathrm{SD}-E} = G(X_{\mathrm{SD}}) \tag{1}$$

where X_{SD} denotes the single-source data input to the model, and through the generator $G(.)$, the extended data $X_{\mathrm{SD}-E}$ are obtained for realizing further classification tasks

$$L_1, L_2, \ldots, L_n = D(X_{\mathrm{SD}}, X_{\mathrm{SD}-E}). \tag{2}$$

The multisource data are passed through the discriminator $D(.)$ to obtain $L_1, L_2, \ldots, L_n$, which represent different losses for different purposes (e.g., distance loss and contrast-adversarial loss), respectively,

$$\alpha = \min(L_1, L_2, \ldots, L_n). \tag{3}$$

By minimizing the combination loss, it allows the model to learn more domain-invariant knowledge and interdomain relationships, thus achieving excellent cross-scene classification.

B. Framework Overview

Assumption $X = \{x_i\}_{i=1}^{N} \epsilon R^d$ is the data from training domain (X_{SD} represents the data from SD, X_{ED} represents the data from ED, and X_{BD} represents the data from BD), and $Y = \{y_i\}_{i=1}^{N}$ is the label corresponding to that data. Here, d and N denote the dimension of the data ($d = 3$ in this task) and the number of source samples, respectively. The proposed TFTnet consists of a generator and a discriminator, as shown in Fig. 1, the X_{SD} data are input into the FDE module to obtain the mean value of amplitude in the X_{SD} domain, and the X_{ED} is obtained by means of self-linear interpolation. X_{BD} is obtained by constraining the X_{SD} and X_{ED} and adding certain Gaussian noise through the MIF module. X_{SD}, X_{ED}, and X_{BD} are input into the discriminator, respectively, and through the AFT module with weight sharing, synergistic information that contains partial frequency domain expressions is obtained. And then, input $F_e(.)$, the classification header C is used to compute the cross-entropy loss, and the projection header P is used to compute the contrast and confrontation loss.

It is important to highlight that, to fully leverage the integrality of HSI spatial and spectral information, dimensional experiments are conducted in this study. As shown in Fig. 2, frequency domain transformations in TFTnet are tested under both 2-D and 3-D conditions. The results indicate that 3-D frequency domain transformations offer a greater advantage in representing HSI information. Moreover, under 3-D conditions, the frequency domain transformation of HSI, as a whole, can more comprehensively utilize the completeness of HSI information.

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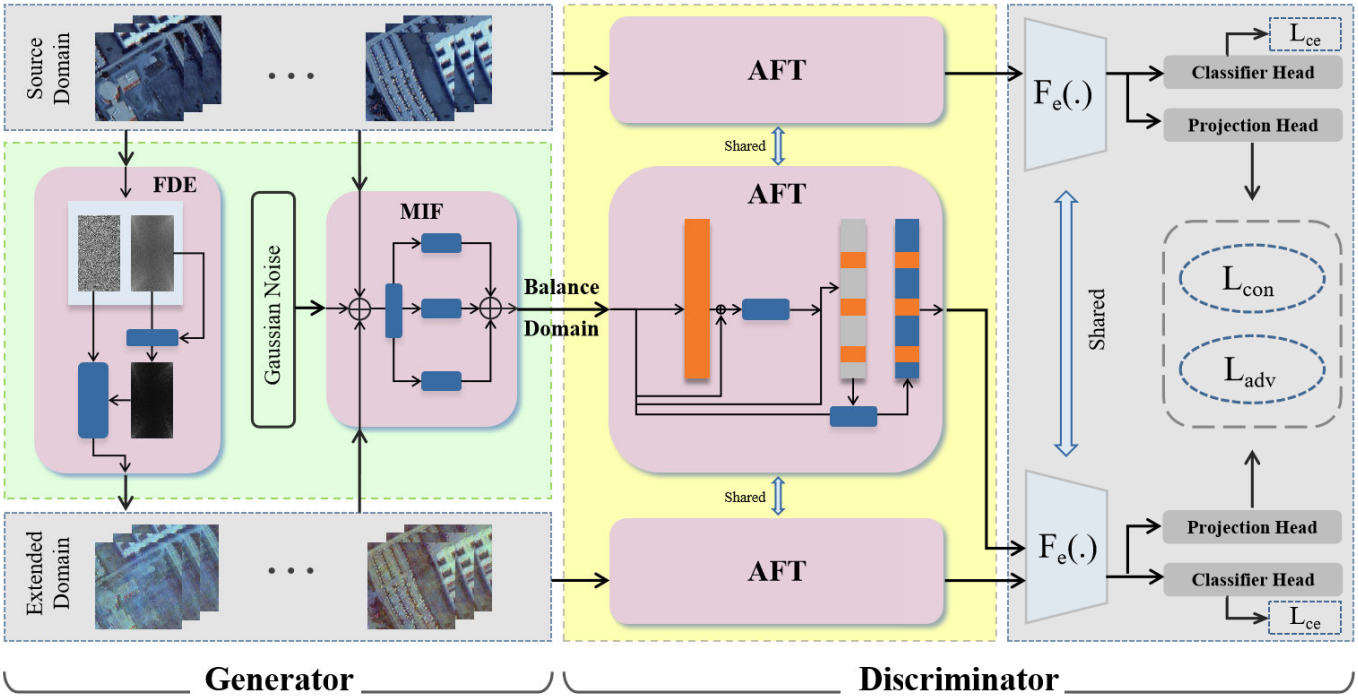


Fig. 1. Proposed TFTnet consists of a generator and a discriminator. The generator comprises an FDE module and an MIF module; the discriminator incorporates a set of weight-sharing AFT modules. $F_e(.)$ represents the feature extractor. L_{ce} represents the classification loss, L_{con} represents the contrast loss, and L_{adv} represents the confrontation loss.

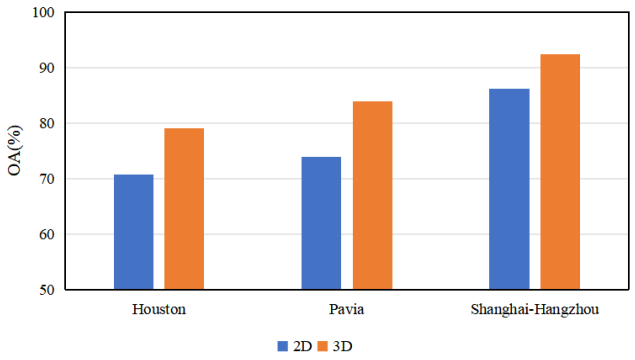


Fig. 2. OA (%) of TFTnet with frequency domain transformations across different dimensions.

C. Interactive Frequency Domain Transformation Generator

Expanding the existing SD effectively is key to achieving single-source DG tasks. Therefore, this article designs a generator G in TFTnet to generate extended samples X_{ED} and X_{BD} with certain domain shifts for increasing the model training sample size. Denote HSI x as X_{SD} data, and its frequency domain transform is

$$F_f(x)(u,v,w) = \sum_{h=0}^{H-1}\sum_{w=0}^{W-1}\sum_{c=0}^{C-1} x(h,w,c) * e^{-j2\pi\left(\frac{h}{H}u+\frac{w}{W}v+\frac{c}{C}w\right)} \tag{4}$$

where $F_f(x)(u,v,w)$ denotes the frequency domain space of $x(h,w,c)$ after a 3-D Fourier transform.

1) FDE Module: Inspired by the semantic preservation property of Fourier phase components and the better cross-DG ability of models highlighting phase information, different from mixing cross-domain sample information, this article designs a new 3-D Fourier-based data augmentation module by mixing in-domain amplitude information to realize in-domain information interaction. X_{SD} is obtained by 3-D Fourier transform of $F_f(x)(u,v,w)$, whose phase and amplitude can be expressed as follows:

$$\begin{aligned} A(x)(u,v,w) &= \left[R^2(x)(u,v,w) + I^2(x)(u,v,w)\right]^{\frac{1}{2}} \\ P(x)(u,v,w) &= \arctan\left[\frac{I(X)(u,v,w)}{R(x)(u,v,w)}\right] \end{aligned} \tag{5}$$

where $R(x)$ and $I(x)$ represent the real and imaginary parts of $F_f(x)$, respectively. For HSI, the Fourier transforms of all channels are computed to obtain the corresponding magnitude and phase information.

Taking advantage of the semantic preservation properties of the Fourier phase components, this article develops models that specifically highlight phase information, which are expected to have better cross-DG capabilities. To achieve this goal, a common choice is to interfere with the amplitude information in the original image through some form of data augmentation. As shown in Fig. 3, we devise a new data augmentation strategy by linearly interpolating between the in-domain amplitude of a single X_{SD} and its own amplitude mean value

$$\widehat{A}\left(x_i^k\right) = \lambda A\left(x_i^k\right) + (1-\lambda)\overline{A}\left(x_i^k\right) \tag{6}$$

where $A(x_i^k)$ represents the intradomain amplitude information, $\overline{A}(x_i^k)$ denotes the mean value of its own amplitude, and $\lambda \sim U(0,\eta)$, where the strength of the enhancement is controlled by the hyperparameter η. The amplitudes after in-domain mixing are combined with the original phase to

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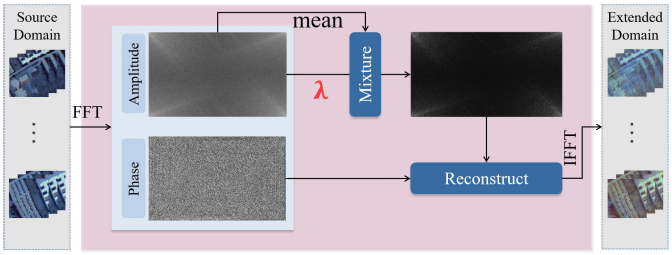


Fig. 3. FDE module. FFT represents 3-D fast Fourier transform, IFFT represents inverse 3-D fast Fourier transform, mean represents the mean amplitude, and λ is used to control the mixing intensity.

form a new Fourier representation

$$F_f\left(\widehat{x_i^k}\right)(u,v,w) = \widehat{A}\left(x_i^k\right)(u,v,w) * e^{-j*P\left(x_i^k\right)(u,v,w)} \tag{7}$$

which is then subjected to an inverse Fourier transform to obtain the enhanced image X_{ED}

$$x_{i,\mathrm{ed}}^k = F_f^{-1}\left[F_f\left(\widehat{x_i^k}\right)(u,v,w)\right] \tag{8}$$

where $F_f^{-1}(x)$ denotes the Fourier inverse transform and $x_{i,\mathrm{ed}}^k$ is the ED X_{ED} with frequency-domain features obtained by X_{SD} after the frequency-domain enhancement module.

Interdomain amplitude swapping aims to reorganize the graph amplitudes in different SDs by swapping them two by two. However, interdomain swapping is not applicable in the HSI single-source cross-scene classification task, so this article proposes intradomain frequency domain mixing for the unique cross-scene task setting of HSI. The mixing intensity is still controlled by the hyperparameters λ. The difference is that the mixing strategy proposed in this article is more adapted to the HSI classification task by doing the mixing through a single SD with its own amplitude mean.

2) MIF Module: The DG task aims to learn to eliminate the interdomain bias problem, i.e., to train the model's generalization ability for unknown samples from a certain domain bias. Its robustness relies on the interdomain bias, which is especially obvious in the HSI cross-scene classification task, where the domain bias between the X_{ED} and the SD is an important factor affecting the model's generalization ability.

Thus, in order to avoid excessive domain shifts between X_{SD} and X_{ED}, an effective approach is to autonomously learn X_{SD} and X_{ED} features to generate X_{BD} with certain joint features, which, in turn, can close the distance and realize the interdomain constraints when the two domains generate excessive domain shifts. It is worth noting that in this article, random Gaussian noise $g = (1/\sqrt{2\pi}\sigma) * e^{-((x-\mu)^2/2\sigma^2)}$ (μ and σ denote the mean and variance, respectively) is added in the process of generating X_{BD}, which ensures the validity of the X_{BD} and makes itself have certain unique features, thus improving the generalization ability of the model for unknown samples.

In this article, the interdomain constraint is realized by weighting based on attention

$$X_{\mathrm{BD}} = w_1 * X_{\mathrm{SD}} + w_2 * X_{\mathrm{ED}} + w_3 * g \tag{9}$$

where w_1, w_2, and w_3 are obtained by means of attention and the values are all in the range of (0, 1), as shown in Fig. 4.

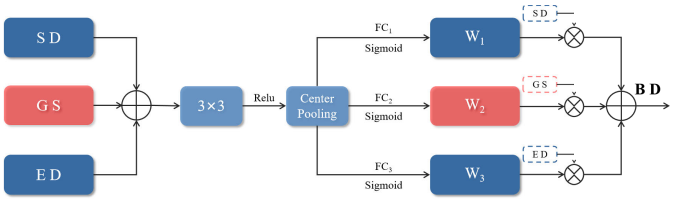


Fig. 4. MIF module. GS represents the Gaussian noise.

X_{SD}, X_{ED}, and g are summed to obtain the mixed input, which passes through a convolutional layer with a kernel size of 3×3, keeping the number of channels unchanged, and does central pooling after passing through the activation function Relu (considering that the HSI classification task is a pixel-level classification, the central pooling can better obtain the key information) and then passes through the respective three fully connected layers and then does Sigmoid to obtain the weights representing X_{SD}, X_{ED}, and g, respectively, which are linearly weighted to obtain X_{BD}.

This article adopts the autonomous learning method to generate X_{BD}, which, on the one hand, eliminates the unknowns brought by random weights, and, at the same time, ensures the reliability of X_{BD} for pulling in the distance of X_{SD} and X_{ED}. Also, this article considers the validity of X_{BD}, i.e., whether it has domain-specific features, so as to improve the generalization performance of the model, and for this reason, Gaussian noise is added in the process of generating X_{BD}.

D. Spatial-Frequency Adaptive Discriminator

The majority of the recent HSI cross-domain classification tasks learn spatial and spectral features of an image in the expectation of achieving better model generalization ability. Frequency domain is also an important way to express image information, and the expressive ability of frequency domain information is more stable, i.e., less sensitive, compared with the spatial–spectral information when facing the problem of domain shift due to interdomain discrepancies. For this reason, this article combines the rich semantic information of HSI and its frequency domain features and expresses the image features of HSI through the synergistic work of the spatial spectrum information and the frequency domain information.

The discriminator D in this article incorporates the AFT module with shared weights and the feature extractor $F_e(.)$ (consisting of a stack of two identical blocks with Conv2D-Relu-MaxPool2D, the convolutional kernel size is 3×3), as well as the classification head C for categorization and the projection head P for training comparisons against loss. X_{SD}, X_{ED}, and X_{BD} are taken as inputs to the discriminator. Also, the joint features, which contains the spatial spectrum information and the frequency domain information, are obtained through the AFT module used as inputs to $F_e(.)$. After $F_e(.)$, C is used for the classification task to train the classification loss (cross-entropy loss), and P is used to train the domain-invariant representation of the learned categories for the contrast-adversarial loss.

1) AFT Module: This module aims to learn autonomously by means of threshold screening to obtain joint expression and realize spatial-frequency synergy. $X_{\mathrm{com}}\{X_{\mathrm{SD}}, X_{\mathrm{ED}}, X_{\mathrm{BD}}\}$,

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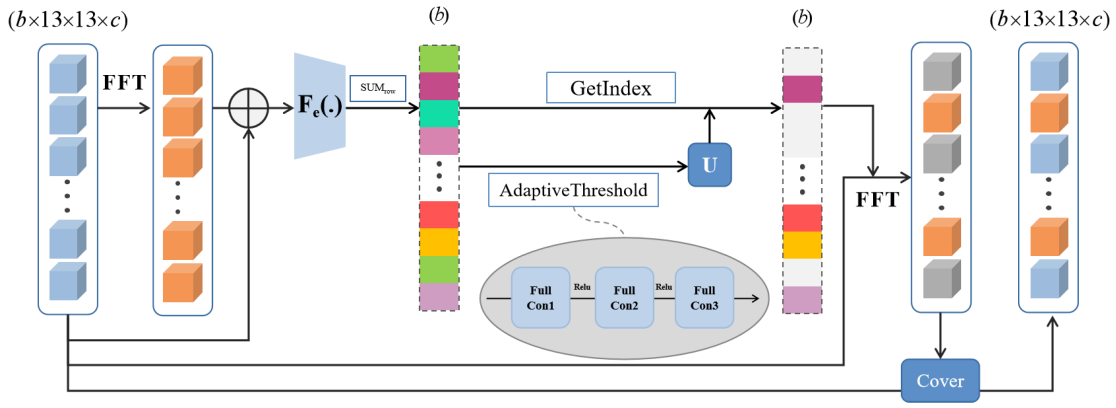


Fig. 5. AFT module.

representing X_{SD}, X_{ED}, and X_{BD} are used as inputs into the discriminator, respectively. As shown in Fig. 5, X_{com} undergoes a 3-D Fourier transform and then does an addition operation with itself

$$X_{\mathrm{com,fft}} = \mathrm{Add}(X_{\mathrm{com}} + F_f(X_{\mathrm{com}})). \tag{10}$$

The obtained mixed data $X_{\mathrm{com,fft}}$ are input to $F_e(.)$, and the obtained feature vectors are accumulated row by row

$$V_{(b,1)} = \mathrm{SUM}_{\mathrm{row}}(F_e(X_{\mathrm{com,fft}})). \tag{11}$$

The obtained output $V_{(b,1)}$ is a 1-D vector of dimension b, which can be regarded as the quantization of the input b patch blocks. After the index acquisition module, the frequency domain indexes are acquired based on the adaptive thresholds

$$\mathrm{Th} = V_{\min} + U * (V_{\max} - V_{\min}). \tag{12}$$

U is used to control the degree of floating of the threshold Th, which is obtained by self-learning, and this method ensures the information interaction of patch blocks while determining the screening range. As shown in Fig. 3, after obtaining the input data, a 1×1 vector is obtained after three fully connected layers, and then, U is obtained after Sigmoid, which, in turn, is filtered V by thresholding Th to obtain the index table

$$\mathrm{Ind}\{i\} = \{i | V[i] \leq \mathrm{Th}\} \tag{13}$$

where $\mathrm{Ind}\{i\}$ is the indexed table, $i \epsilon (0, b)$. Then, based on the indexed table, the filtered portion X_{com} is transformed in the frequency domain and covered with X_{com}

$$X_{\mathrm{com,sel\text{-}fft}} = \mathrm{Cover}(F_f(X_{\mathrm{com}}(\mathrm{Ind}\{i\}) \rightarrow X_{\mathrm{com}}). \tag{14}$$

At this point, $X_{\mathrm{com,sel\text{-}fft}}$ is obtained as the joint feature, in which the spatial spectrum information and the frequency domain information together express the HSI features.

This joint information is input into $F_e(.)$, and the HSI cross-domain classification is realized through C and P.

2) Training Loss: In TFTnet, the optimization of G and D is achieved through cross-entropy loss and contrast-adversarial loss.

X_{SD}, X_{ED}, and X_{BD} have the same labeling space, so based on the cross-entropy loss, the categorization loss of X_{com} can be defined as follows:

$$L_{\mathrm{com}}(X_{\mathrm{com}}, Y) = \frac{1}{N} \sum_i L_{\mathrm{ce}}(p(x_{\mathrm{com},i}, y_i)) \tag{15}$$

where $L_{\mathrm{com}}(X_{\mathrm{com}}, Y)$ is equivalent to X_{SD}, X_{ED}, and X_{BD} classification loss. To prevent excessive separation and aggregation within and between domains, supervised contrastive adversarial learning is introduced to motivate D to learn domain-invariant representations

$$\begin{aligned} L_{\mathrm{con}} &= -\sum_{i=0}^{N} \frac{1}{|P(i)|} \sum_{p \epsilon P(i)} \log \frac{\exp\left(S\left(Z_i, Z_p^+\right)/\tau\right)}{\sum_{a \epsilon A(i)} \exp\left(S\left(Z_i, Z_a^-\right)/\tau\right)} \\ L_{\mathrm{adv}} &= -\sum_{c} \sum_{i=0}^{n_c} \frac{1}{|P_c(i)|} \sum_{p \epsilon P_c(i)} \\ &\quad \times \log \frac{\exp\left(S\left(Z_i, Z_p^+\right)/\tau\right)}{\sum_{a \epsilon A_c(i)} \exp\left(S\left(Z_i, Z_a^-\right)/\tau\right)} \end{aligned} \tag{16}$$

where L_{con} represents the contrast loss, which makes the samples of the same class closer together and the samples of different classes farther apart; i.e., it realizes the model to learn a shared representation of the samples of the same class, and L_{adv} represents the confrontation loss, which ensures that the ED has domain-specific information.

The optimization process of D includes the optimization of the combination of $F_e(\theta_f)$, $C(\theta_c)$, and $P(\theta_p)$

$$\min_{\theta_f, \theta_c, \theta_p} L_D = L_{\mathrm{com,SD}} + L_{\mathrm{com,ED}} + L_{\mathrm{com,BD}} + \lambda_1 L_{\mathrm{con}} \tag{17}$$

where λ_1 is the hyperparameter used to balance the supervised comparison loss. The optimization process for G consists of optimizing $G(\theta_g)$, using the X_{ED} classification loss as an auxiliary

$$\min_{\theta_g} L_D = L_{\mathrm{com,ED}} + \lambda_2 L_{\mathrm{adv}} \tag{18}$$

where λ_2 is the hyperparameter used to balance against loss.

III. Experiments

A. Datasets

The experiments are based on three cross-scene HSI datasets, namely, the Houston, Pavia, and Shanghai-Hangzhou

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TABLE I
NUMBER OF SOURCE AND TARGET SAMPLES FOR THE HOUSTON DATASET

No.	Class	Houston 2013 (SD)	Houston 2018 (TD)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
	Total	2530	53200

datasets. TFTnet is implemented on the PyTorch [39] platform, where a 10G GeForce RTX 2080 Ti is the base hardware platform.

1) Houston Dataset: The dataset consists of scenes from Houston and Debes et al. [40] by Le Saux et al. [41], which were collected from various sensors on the University of Houston campus and in the community. The Houston 2013 [40] dataset has an image resolution of 2.5 m, 349 × 1905 pixels, 144 spectral bands, and a wavelength range of 380–1050 nm. The Houston 2018 [41] dataset includes 48 spectral bands and the same wavelength range, and the image has a spatial resolution of 1 m. There are seven invariant classes in the scene. In this article, an overlapping area of 209 × 955 is selected to extract 48 spectral bands (wavelength range 0.38–1.05 μm) from the Houston 2018 scene. The sample categories and number of samples are shown in Table I. The pseudocolor and true value maps are shown in Fig. 6.

2) Pavia Dataset: The University of Pavia and Pavia Center [42], [43] datasets were acquired by the Reflectance Optical System Imaging Spectrometer (ROSIS) sensor. The sensor has a total of 115 bands with a spectral range of 430–860 nm. After processing, the data from the University of Pavia have 103 bands, and the data from the Pavia Center have 102 bands. In this article, the 102 bands common to both data are used. The University of Pavia has a size of 610 × 340, and the Pavia Center has a size of 1096 × 715, with a spatial resolution of 1.3 m. Both images have nine classes of ground objects. In this article, seven categories shared by both data are used. The specific number of samples is shown in Table II, and the pseudocolor and true value maps are shown in Fig. 7.

3) Shanghai-Hangzhou Dataset: The Shanghai-Hangzhou dataset was captured by the EO-1 [44] Hebron Ion Hyperspectral Sensor, which retains 198 bands. The Shanghai data image has 1600 × 230 pixel points. The Hangzhou data image has 590 × 230 pixel points. The linear distance between Shanghai and Hangzhou is 162 km. Table III lists the three common land cover classes used for cross-scene classification, and their pseudocolor and truth maps are shown in Fig. 8.

B. Evaluation Metrics

In this article, overall accuracy (OA), average accuracy (AA), and kappa coefficient (Ka) are used to evaluate the model classification performance [45].

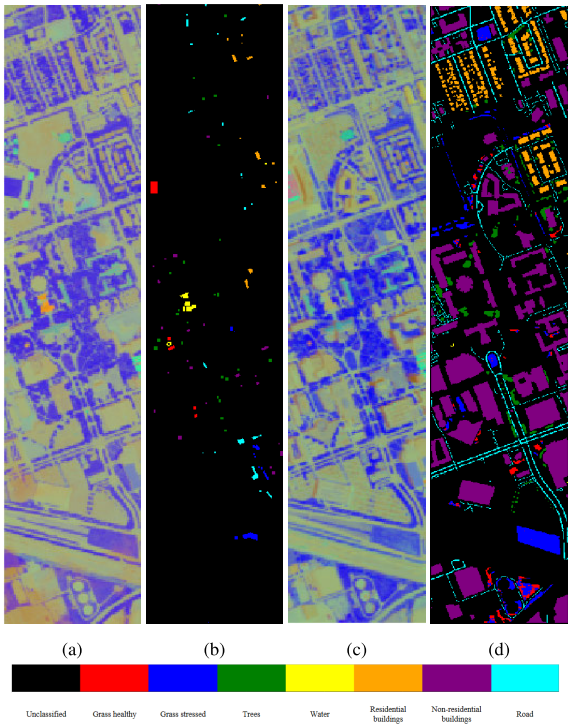


Fig. 6. Pseudocolor image and ground-truth map of the Houston dataset. (a) Pseudocolor image of Houston 2013. (b) Ground-truth map of Houston 2013. (c) Pseudocolor image of Houston 2018. (d) Ground-truth map of Houston 2018.

TABLE II
NUMBER OF SOURCE AND TARGET SAMPLES FOR THE PAVIA DATASET

No.	Class	PaviaU (SD)	PaviaC (TD)
1	Tree	3064	7598
2	Asphalt	6631	9248
3	Brick	3682	2685
4	Bitumen	1330	7287
5	Shadow	947	2863
6	Meadow	18649	3090
7	Bare soil	5029	6584
	Total	39332	39355

TABLE III
NUMBER OF SOURCE AND TARGET SAMPLES FOR THE SHANGHAI-HANGZHOU DATASET

No.	Class	Hangzhou (SD)	Shanghai (TD)
1	Water	18043	123123
2	Land/Building	77450	161689
3	Plant	40207	83188
	Total	135700	368000

OA indicates the number of correctly classified samples divided by the total number of samples, which shows the probability that all samples are correctly classified in general and takes the value of 0–100 (%); AA refers to the average value obtained by summing the classification accuracies of each feature and takes the value of 0–100 (%); and Ka is a statistical measure of consistency. For classification problems,

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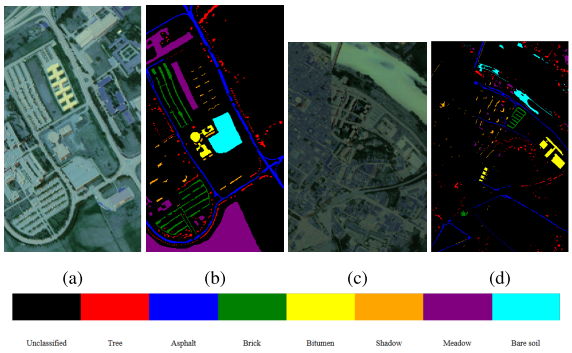


Fig. 7. Pseudocolor image and ground-truth map of Pavia dataset. (a) Pseudocolor image of University of Pavia. (b) Ground-truth map of University of Pavia. (c) Pseudocolor image of Pavia Center. (d) Ground-truth map of Pavia Center.

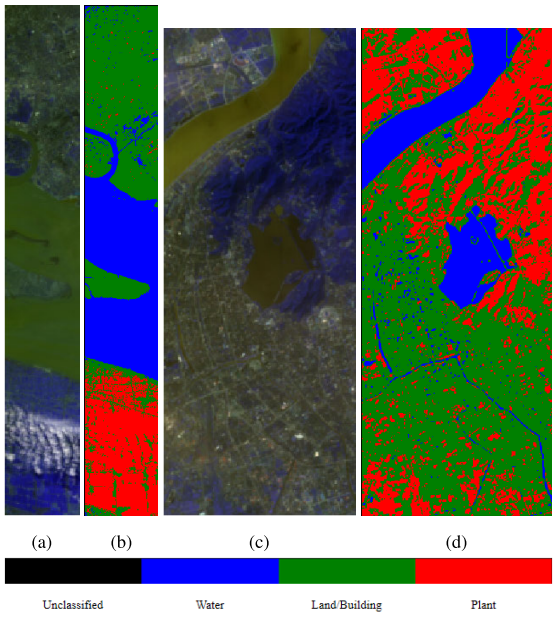


Fig. 8. Pseudocolor image and ground-truth map of Shanghai-Hangzhou dataset. (a) Pseudocolor image of Shanghai. (b) Ground-truth map of Shanghai. (c) Pseudocolor image of Hangzhou. (d) Ground-truth map of Hangzhou.

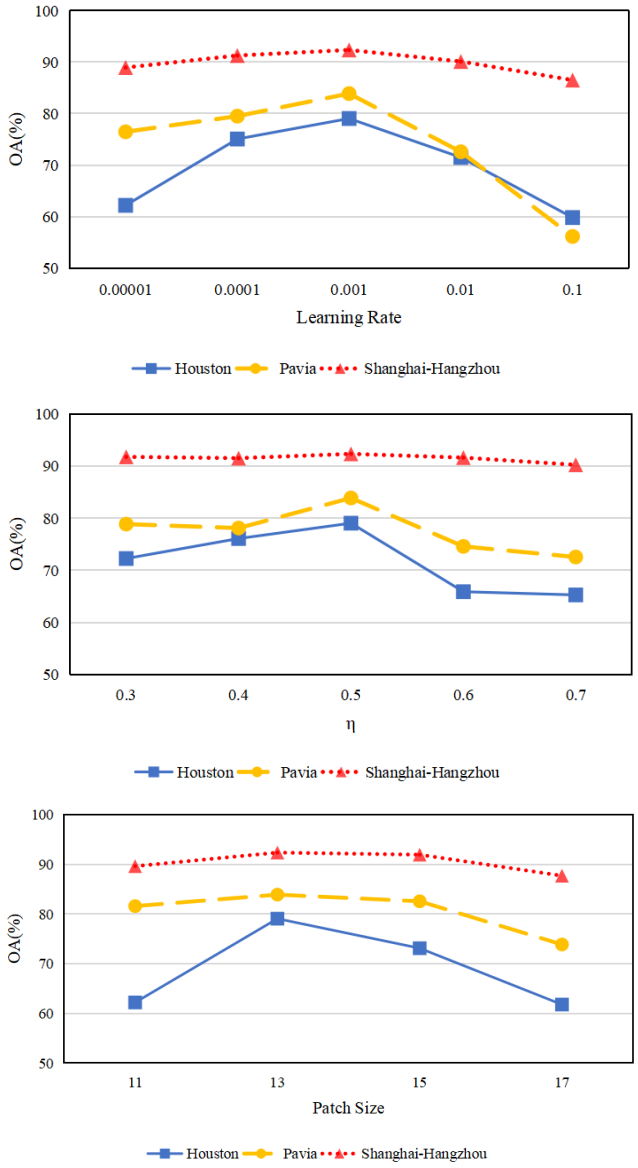


Fig. 9. OA (%) of different parameter selections on the three datasets.

in which the higher the value of Ka, represents the higher degree of consistency, the value is in $[-1, 1]$, usually greater than 0 in which the higher the value of Ka, represents the higher degree of consistency, in order to balance the numerical relationship between the three indicators; this article will be all the Ka that take the value of $\times 100$. In this article, these three different evaluation indexes are used to evaluate TFTnet together.

C. Parametric Experiment

In order to evaluate the generalizability of TFTnet for the three TDs, certain parametric experiments are conducted in this article using the Adam optimizer. Batch size values of 256 on the Pavia dataset, the Houston dataset, and the Shanghai-Hangzhou dataset are used to control the hyperparameters of the enhancement strengths in the frequency domain, the overall learning rate of the model, and the batch size patch, that are taken from $\{1e-5, 1e-4, 1e-3, 1e-2, 1e-1\}$, $\{0.3, 0.4, 0.5, 0.6, 0.7\}$, and $\{11 \times 11, 13 \times 13, 15 \times 15, 17 \times 17\}$, respectively. This article uses OA metrics for quantitative research.

During the gradient update process, the gradient of the loss function is used to estimate the hyperparameters of the model weights after adjusting by the learning rate. Fig. 9 provides the classification results of the three datasets with different base learning rates, different enhancement parameters η, and different patch sizes, corresponding to the optimal base learning rates of $1e-3$, 0.5, and 13×13.

For the hyperparameters λ_1 and λ_2, which are used to balance the supervised comparison and the adversarial loss in optimizing the weighted loss of C and D, the default setting in the experiments is 1.

D. Comparison Experiment

In order to demonstrate the superior performance of TFTnet in DG tasks, some DG methods are selected as comparison

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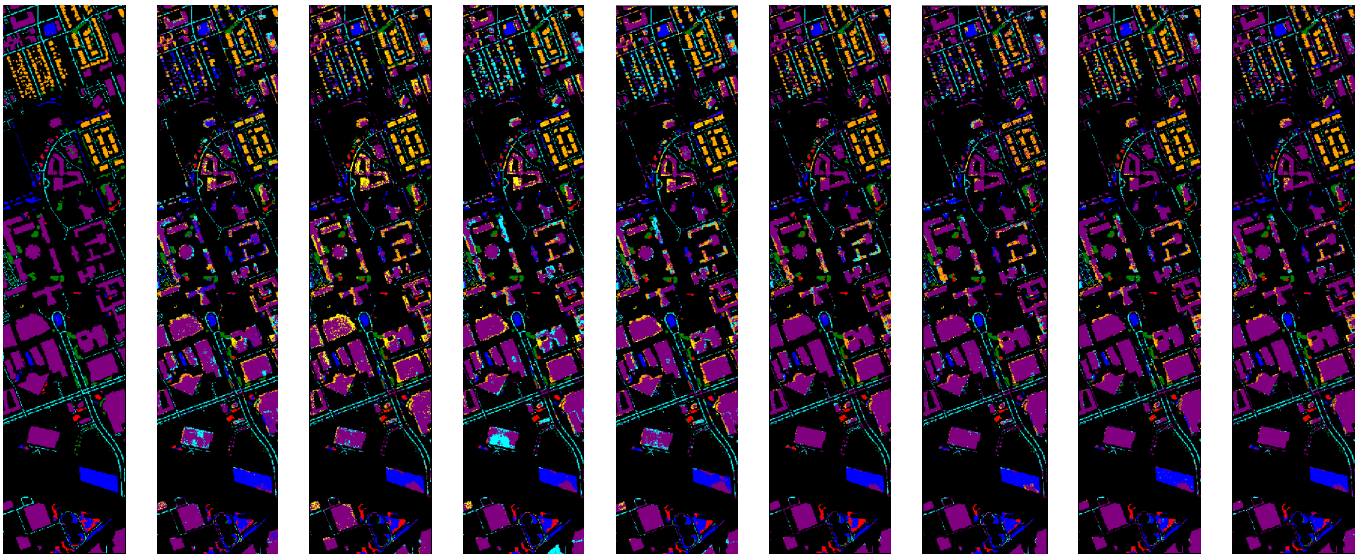


Fig. 10. Visualization and classification maps of different models on the Houston dataset (TD: Houston 2018). H18-GT represents the ground-truth map of Houston 2018.

experiments for validating in this article. Among them, PDEN [46], LDSDG [21], SagNet [47], SDEnet [29], LLURnet [30], and S2ECnet [32] are DG methods. In the DG method, only labeled SD data (80% for training and 20% for validation) are used. Among them, due to the restriction of ResNet18 on input images, the patch size of PDEN, LDSDG, and SagNet was set to 32, respectively. $\{1e-5, 1e-4, 1e-3, 1e-2, 1e-1\}$ and $\{1e-3, 1e-2, 1e-1, 1e+0, 1e+1, 1e+2\}$ were selected as the optimal base learning rate and regularization parameters for each comparison algorithm, and the corresponding optimal parameters were found through cross validation. For the Houston dataset, Houston 2013 is selected as SD and Houston 2018 as TD. For the Pavia dataset, Pavia University is selected as SD and Pavia Center as TD. For the Shanghai-Hangzhou dataset, Hangzhou is selected as SD, and Shanghai was selected as TD. Tables IV–VI show the results for various categories of OA, AA, and Kappa on the Houston, Pavia, and Shanghai-Hangzhou datasets, respectively, with the best results retained.

The TFTnet proposed in this article obtains the best evaluation metrics in all datasets, compared with other DG methods, with different degrees of improvement in all three evaluation metrics. Notably, compared with the mainstream DG models SDEnet, LLURnet, and S2ECnet currently applied to the HSI cross-scene classification task, the three datasets show an improvement of 0.42%–4.05% for OA, 0.14%–1.96% for AA, and 0.59%–4.73% for Kappa, which effectively demonstrates the effectiveness of this article's method due to the incorporation of the learning of frequency domain information, and it shows stronger adaptability when facing the domain shift problem and also proves the effectiveness of this article's method. Figs. 10–12 show the visual classification results of different models on the three datasets.

To show the computational complexity of the different methods, Table VII lists one epoch training time for all experimental data. All experiments were performed on the same hardware platform (GeForce RTX 2080 Ti). It can be seen that the computational cost of TFTnet is much lower than the other compared methods.

This is due to the fact that only two layers of Conv2D-Relu-MaxPool2D blocks are used to learn the domain-invariant representation, which is much simpler than the DG methods that use VGG or ResNet as the backbone. In addition, the unique frequency-domain augmentation block does not introduce additional computational cost to the model, and the frequency-domain augmentation-based generator has a simpler structure with lower complexity than other DG models.

E. Generalization Performance

This article conducts t-SNE visualization experiments on three different datasets, visually expressing the intraclass and interclass aggregation effects of different samples through clustering diagrams.

As shown in Fig. 13(c), the feature distributions of ED and SD exhibit high similarity in interclass relationships, indicating the reliability of ED. However, ED shows greater dispersion in intraclass features, suggesting a certain offset between the ED and SD samples of the same class, which confirms the effectiveness of ED.

As shown in Fig. 13(d), without the AFT module, the discriminator exhibits more scattered points in the classification map of TD. Moreover, its intraclass aggregation and interclass separation effects are poorer compared with Fig. 13(g). This indicates that integrating the AFT module into the model allows it to learn more frequency domain stability. Consequently, when facing cross-scene tasks, the model demonstrates better generalization capability.

It is worth noting that, as shown in Fig. 13(e), the clustering effect of TFTnet based on 2-D Fourier transform is inferior to that based on 3-D Fourier transform. This suggests that the 3-D Fourier transform is more suitable for HSI data features.

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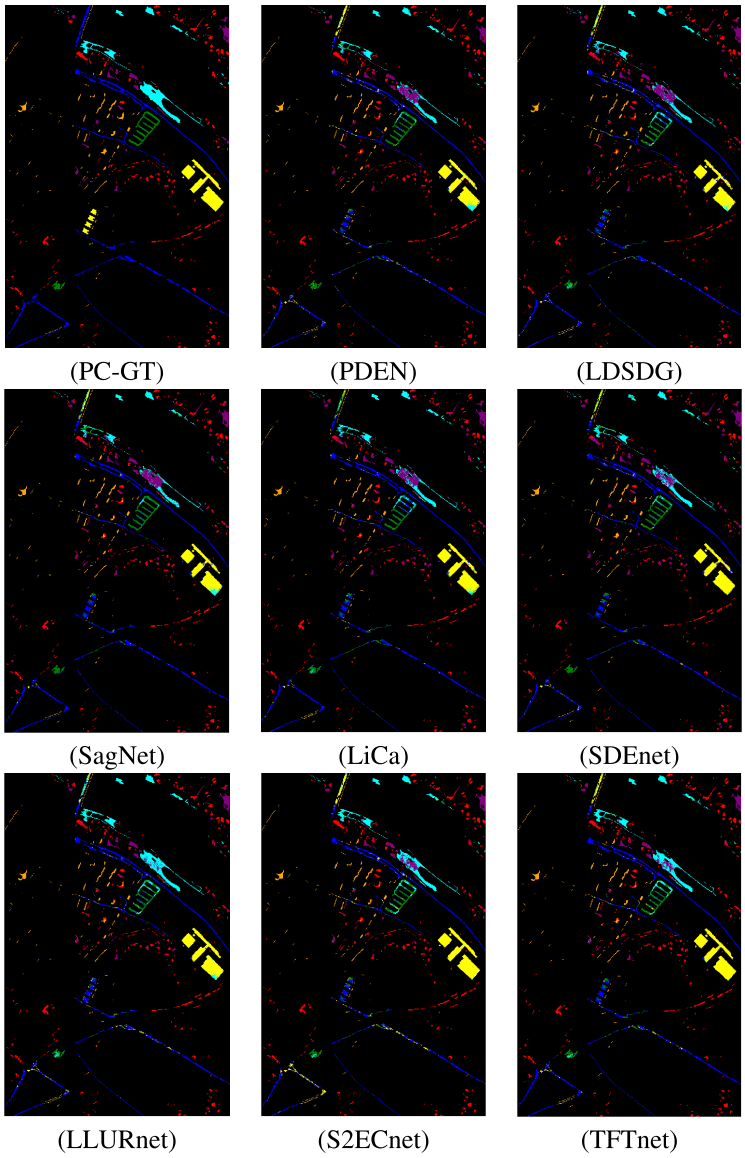


Fig. 11. Visualization and classification maps of different models on the Pavia dataset (TD: Pavia Center). PC-GT represents the ground-truth map of Pavia Center.

TABLE IV
CLASS-SPECIFIC AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE TARGET SCENE HOUSTON 2018 DATA

Class	PDEN [46]	LDSDG [21]	SagNet [47]	LiCa [28]	SDEnet [29]	LLURnet [30]	S2ECnet [32]	TFTnet
1	37.95	37.57	27.63	36.23	64.23	56.91	61.57	56.47
2	51.24	57.78	54.36	46.05	84.74	80.63	72.07	77.8
3	45.26	47.98	47.4	42.31	41.87	66.92	57.52	59.33
4	77.16	63.45	86.48	76.67	100	100	100	100
5	53.91	57.17	55.92	54.52	62.88	76.25	57.66	66.9
6	87.27	83.92	90.26	90.05	88.4	78.11	91	89.65
7	26.73	24.71	17.14	18.84	43.69	61.74	47.1	49.36
OA	69.83	68.84	70.74	67.79	77.12	75.09	78.17	**79.04**
AA	54.22	53.23	54.17	58.21	69.4	70.37	69.56	**71.36**
Ka	47.89	48.02	46.54	43.19	60.71	60.57	61.75	**63.71**

Fig. 13(f) and (g), respectively, depicts the clustering results of the baseline model [29] and TFTnet. Compared with the baseline, TFTnet shows better intraclass and interclass relationships, demonstrating TFTnet's stronger generalization performance in HSI cross-scene classification tasks.

F. Ablation Experiment

The data enhancement module based on the frequency domain transform, the weighted constraint module for eliminating excessive interdomain variability together form the generator, and the AFT module for realizing the

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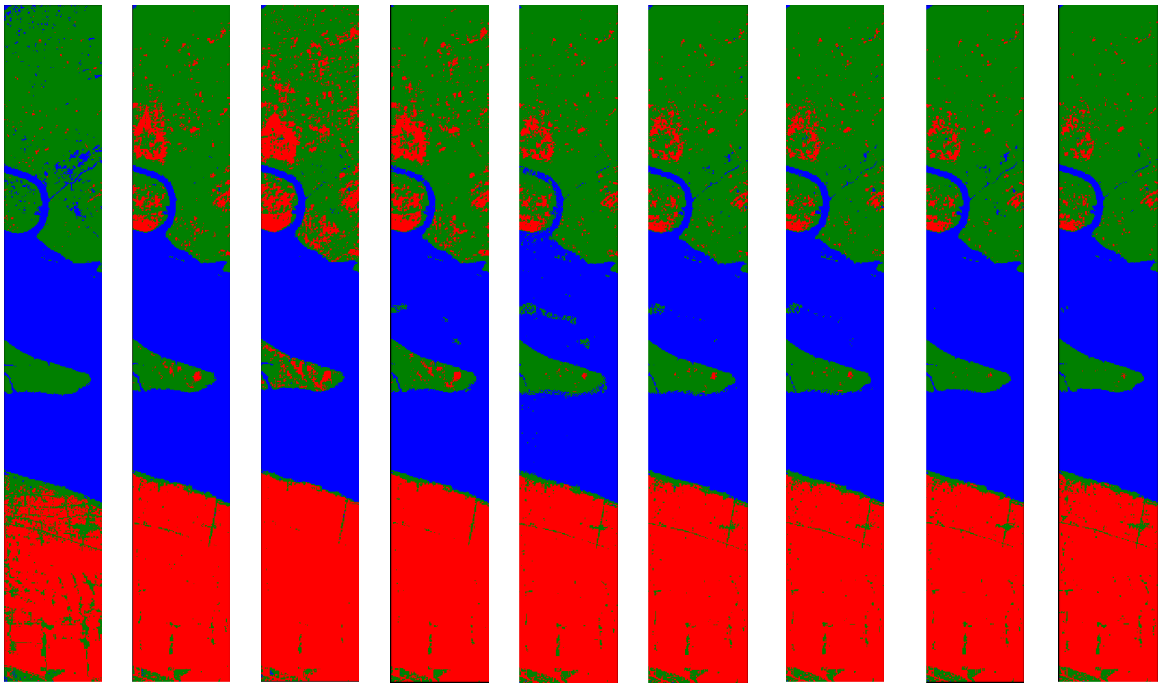


Fig. 12. Visualization and classification maps of different models on the Shanghai-Hangzhou dataset (TD: Shanghai). SH-GT represents the ground-truth map of Shanghai.

TABLE V

CLASS-SPECIFIC AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE TARGET SCENE PAVIA CENTER DATA

Class	PDEN [46]	LDSDG [21]	SagNet [47]	LiCa [28]	SDEnet [29]	LLURnet [30]	S2ECnet [32]	TFTnet
1	65.59	91.64	93.88	73.54	89.42	86.6	95.31	95.33
2	83.45	52.19	58.67	66.14	84.96	86.31	80.48	81.56
3	31.92	8.66	10.35	39.14	79.96	69.16	65.81	69.16
4	74.66	82.87	81.83	58.56	83.11	77.86	84.7	85.98
5	65.01	67.83	91.44	90.46	93.75	87.91	93.47	90.25
6	55.86	63.08	70.29	53.56	71.26	72.59	68.8	67.86
7	64.33	71.72	83.64	58.63	55.85	80.32	79.16	82.34
OA	67.56	68.23	73.93	59.03	79.83	81.67	82.93	**83.88**
AA	62.94	62.57	70.01	62.72	79.76	80.11	81.1	**81.78**
Ka	59.1	62.96	68.53	46.07	75.86	77.9	79.51	**80.59**

TABLE VI

CLASS-SPECIFIC AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS FOR THE TARGET SCENE SHANGHAI DATA

Class	PDEN [46]	LDSDG [21]	SagNet [47]	LiCa [28]	SDEnet [29]	LLURnet [30]	S2ECnet [32]	TFTnet
1	83.51	89.54	92.32	86.77	90.59	92.66	95.96	94.78
2	77.78	70.83	71.05	71.55	85.36	84.48	84.75	86.83
3	98.49	99.7	98.27	94.99	99.61	99.75	99.94	99.47
OA	85.01	84.95	86.19	82.58	90.33	90.67	91.93	**92.35**
AA	86.59	86.69	87.21	84.44	91.85	92.3	93.55	**93.69**
Ka	77.37	75.78	79.28	77.14	85.22	85.77	87.71	**88.3**

TABLE VII

EXECUTION TIME (IN SECONDS) OF ONE EPOCH TRAINING IN DIFFERENT METHODS

Model		PDEN [46]	LDSDG [21]	SagNet [47]	LiCa [28]	SDEnet [29]	LLURnet [30]	S2ECnet [32]	TFTnet
Pavia	Training(S)	126.6	315.54	216.35	103.51	62.64	71.77	447.31	**30.6**
Houston	Training(S)	27.02	46.58	102.69	28.79	19.53	23.63	254.58	**11.3**
H-S	Training(S)	197.46	436.14	447.63	207.62	187.11	247.14	1552.31	**167.74**

synergy between the spatial–spectral information and the frequency domain information are the key modules in the discriminator. In order to evaluate the contribution of each module to TFTnet, an ablation experiment is performed by removing each subgroup for the entire framework.

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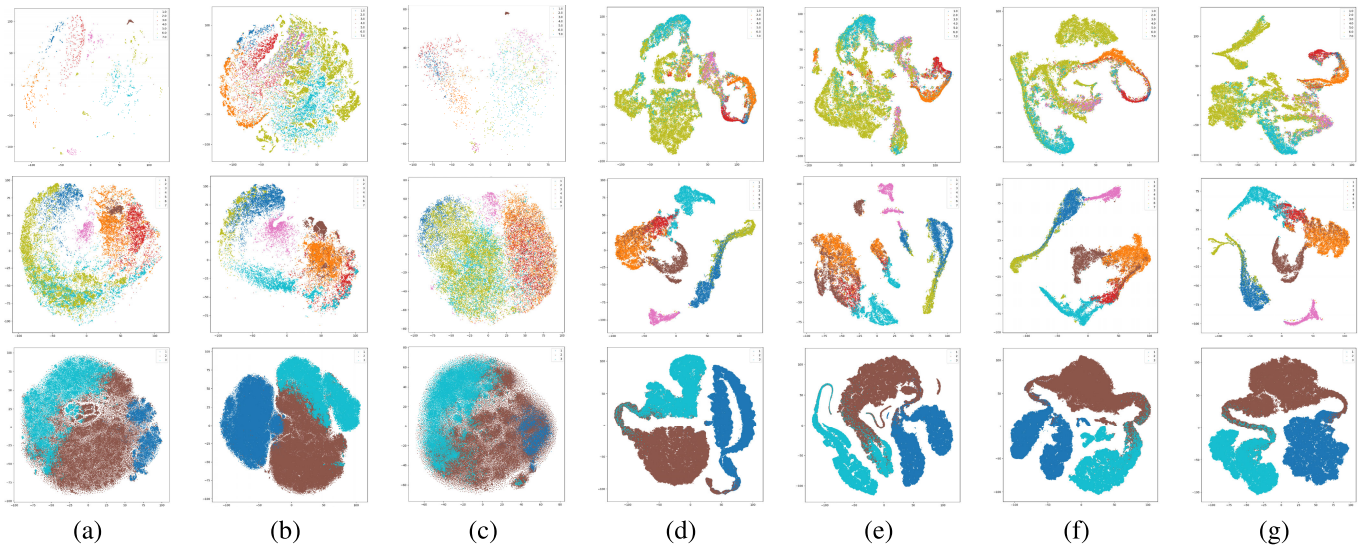


Fig. 13. t-SNE of class separability of three datasets (the first line represents the Houston Dataset, the second line represents the Pavia Dataset, and the third line represents the Shanghai-Hangzhou Dataset). (a) Raw samples of SD. (b) Raw samples of TD. (c) Raw samples of ED. (d) TD features by TFTnet (no AFT module). (e) TD features by TFTnet (2-D Fourier transform). (f) TD features by SDEnet [29]. (g) TD features by TFTnet.

TABLE VIII
ABLATION COMPARISON OF EACH VARIANT OF TFTNET

Model	TFTnet (no FDE)	TFTnet (no MIF)	TFTnet (no AFT)	TFTnet
Dataset	Pavia			
OA	80.48	81.69	81.95	**83.88**
AA	81.04	80.61	80.57	**81.78**
Ka	76.69	77.94	78.32	**80.59**
Dataset	Houston			
OA	77.91	78.06	78.35	**79.04**
AA	69.86	70.5	70.85	**71.36**
Ka	62.04	62.25	62.69	**63.71**
Dataset	Hangzhou-Shanghai			
OA	91.27	91.32	91.41	**92.35**
AA	92.83	92.5	92.45	**93.69**
Ka	86.38	86.51	85.69	**88.3**

Three variants of the framework are present in the ablation experiments, namely, the following hold: 1) TFTnet (no FDE module): the FDE module is replaced by the encoder–decoder structure used to generate the ED in the baseline [29] framework; 2) TFTnet (no MIF module): the MIF module is replaced by the module used to randomly generate the BD in the baseline [18] framework (which includes removing the Gaussian noise addition); and 3) TFTnet (no AFT module): remove the AFT module from the TFTnet framework.

As shown in Table VIII, TFTnet clearly outperforms all the variant frameworks and obtains a large performance improvement, which demonstrates the effectiveness of each module.

In three different datasets, compared with TFTnet, the OA of "no FDE" is reduced by 1.08%–3.4%, which proves the effectiveness of the FDE module in simulating the HSI frequency domain features; the OA of "no MIF" is reduced by 0.98%–1.31%, which proves that the contribution of self-learning weights to generating BD is better than random generation; and the OA of "no AFT" is reduced by 0.69%–1.93%, which proves that although the constructed brand-new generator has a more superior performance, it fails to integrate spatial–spectral and frequency-domain information effectively, thereby underscoring the effectiveness of the spatial-frequency synergistic module.

It is worth noting that the OA interval of HZ-SH on the three variants is only 0.14%, while the OA intervals of Pavia and Houston are 2.09% and 0.44%, respectively. A potential reason for this is the number of training samples; HZ-SH has a large number of training samples (135 700) and only needs to categorize three samples with low correlation (water, land/building, and plant). The large amount of training data makes HZ-SH have better stability in TFTnet, which also proves that the quality of the dataset has some correlation with model stability.

IV. CONCLUSION

In this article, we propose a novel TFTnet for HSI classification tasks, which includes a generator composed of an FDE module and an MIF module for effective sample expansion and a discriminator consisting of an AFT module for integrating spatial–spectral and frequency domain information. The frequency-domain transform-based generator is more suitable for single-source DG tasks, simulates the frequency-domain characteristics of TD, and generates reliable BD to improve the generalization performance of the model. The AFT module uses a self-learning mode to obtain a joint representation of information, achieving spatial-frequency synergy.

In addition, to demonstrate that TFTnet maintains promising classification performance in cross-scene tasks, we conduct comparison experiments and ablation experiments on the Houston, Pavia, and Shanghai-Hangzhou datasets. The results of the former show that our method outperforms the existing methods, while the latter confirms the effectiveness of all modules in performing DG tasks. To visually represent TFTnet's generalization performance, we present the aggregation effects

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of each class through generalization experiments, and the results clearly show the intraclass and interclass aggregation effects for different classes under various conditions.

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Jun Zhang received the B.S. and Ph.D. degrees from Hebei University of Technology, Tianjin, China, in 1999 and 2011, respectively.

He is currently an Associate Professor with the School of Artificial Intelligence, Hebei University of Technology. His research interests include machine learning and intelligent computing.



Cheng Zhang received the B.S. degree in the Internet of Things from Tianjin University of Science and Technology, Tianjin, China, in 2021. He is currently pursuing the M.S. degree with the School of Artificial Intelligence, Hebei University of Technology, Tianjin.

His research interests include machine learning and intelligent computing.



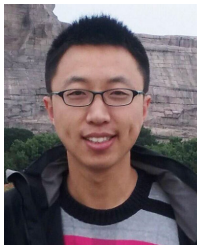
Shuai Liu received the Ph.D. degree from the School of Information and Communication Engineering, Dalian University of Technology (DUT), Dalian, China, in 2023.

He is currently an Instructor with the School of Artificial Intelligence and Data Science, Hebei University of Technology, Tianjin, China. His research interests include object detection, multiobject tracking in common, and aerial scenes.



Zhenwei Shi (Senior Member, IEEE) is currently a Professor and the Dean of the Image Processing Center, School of Astronautics, Beihang University, Beijing, China. He has authored or co-authored over 200 scientific articles in refereed journals and proceedings, including IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), and the IEEE International Conference on Computer Vision (ICCV). His research interests include remote sensing image processing and analysis, computer vision, pattern recognition, and machine learning.

Prof. Shi serves as an Editor for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, *Pattern Recognition*, *ISPRS Journal of Photogrammetry and Remote Sensing*, and *Infrared Physics and Technology*. His personal website is http://levir.buaa.edu.cn/.



Bin Pan (Member, IEEE) received the B.S. and Ph.D. degrees from the School of Astronautics, Beihang University, Beijing, China, in 2013 and 2019, respectively.

Since 2019, he has been an Associate Professor with the School of Statistics and Data Science, Nankai University, Tianjin, China. His research interests include machine learning, remote sensing image processing, and multiobjective optimization.

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